Engineering-Ethics Scaffolding for Deployed Public-Good AI Mental-Health Systems: A Combined License, Deployment-Pause Posture, and Rule-Coverage CI Pattern

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Preprint v1, 2026-04-30. License: CC BY 4.0. Code and reference implementation: https://github.com/wudaming00/stay (test suite under MIT; system prompt under the Stay System Prompt License v1, see `LICENSE-PROMPT.md`). A companion paper proposing a four-utterance-class anti-pattern catalog and a measurement-validity reading of the VERA-MH benchmark is available at the same repository (`docs/paper-B-antipattern-catalog-draft.md`); citations to it appear here as motivation.

Abstract

Tens of millions of users now talk to AI systems about their mental health. OpenAI disclosed in October 2025 that roughly 0.15% of weekly active ChatGPT users (over a million people per week) show explicit suicidal-planning indicators.[1] The regulatory environment shifted accordingly through 2025: an FTC 6(b) inquiry in September, FDA Digital Health Advisory Committee deliberations in November, an APA Health Advisory the same month, and a Florida federal court ruling in *Garcia v. Character Technologies*[2] that denied First-Amendment-defense dismissal of tort claims against an AI-companion company (the case settled in January 2026 with terms undisclosed).

Despite this, no published governance pattern for *public-good* deployed AI mental-health systems survives the open-source fork. Existing patterns split into two camps. Closed commercial systems with internal review (Therabot, Wysa, Limbic Access) keep their governance private. Open-source companion and roleplay systems generally ship without safety-section retention constraints, and the fork-strips-safety failure mode is documented: in 2023 the NEDA Tessa chatbot replaced its scripted helpline with a generative version, then immediately produced eating-disorder-promoting advice and was pulled within days.

This paper proposes a three-tool engineering-ethics governance pattern intended to fill that gap, documents a prior-art search supporting the novelty of the combination, and offers one reference implementation. The three tools are: a restricted-use license on system-prompt text (`LICENSE-PROMPT.md`) that combines Hippocratic-License-style use-restriction, OpenRAIL-style behavioral constraint, and Mozilla-MPL-style file-level copyleft, applied to four named safety-critical sections; a self-imposed deployment-pause posture with auditable trigger conditions, named adjudicators, deadlines, and an author-unavailability dead-man switch; and a CI-enforced rule-coverage check mapping registered safety rules to behavioral test assertions. Each

component has substantial prior art. The combination applied to a deployed public-good AI mental-health system is, by our search, not previously published. Stay (thestay.app) is the reference implementation: two of three components are production-active, the third is API-validated with the operational wiring still to be configured. Stay is the forkable artifact; the pattern is the contribution.

The paper does not claim any of the following: that the package substitutes for clinical evaluation, regulatory compliance, or operator liability insurance; that LICENSE-PROMPT has been tested in court; that any external party has yet adopted the pattern; or that the rule-coverage CI catches semantic regressions in the registered rules.

[1] OpenAI, "Strengthening ChatGPT's responses in sensitive conversations," 2025-10-27. The 0.15% / 0.07% rates are from OpenAI; absolute counts are secondary-source extrapolations against ~800M weekly actives (TechCrunch, *Guardian*, *Wired*, BBC, *BMJ* 391:r2290).

[2] M.D. Fla. 6:24-cv-01903, filed Oct 2024; Conway J. May 20 2025 motion-to-dismiss ruling; settled Jan 7 2026 with terms undisclosed (NYT / CNBC / Guardian / CNN / JURIST).

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1 · Introduction

1.1 The deployment landscape and the governance vacuum

By late 2025 the deployed AI mental-health landscape had bifurcated into two regimes, neither of which carried an externally inspectable governance package.

The first regime is closed-commercial deployment with internal review. Therabot (Heinz et al. 2025, *NEJM AI*) was developed under six years of expert-curated CBT corpus construction and ran as an actively-monitored RCT with human safety review of every flagged exchange. Wysa (Inkster et al. 2018) holds an FDA Breakthrough Device Designation and is integrated into NHS Talking Therapies. Limbic Access (Habicht et al. 2024, *Nature Medicine*) was the first AI mental-health chatbot to gain UKCA Class IIa medical-device status. These systems have substantive internal governance. The governance is not externally inspectable: the system prompts, the safety SOPs, the trigger conditions, and the review processes are not published. A grassroots developer cannot fork Wysa, and Therabot's weights are not public.

The second regime is open-source release without safety-section retention constraints. Character.AI operates under the open-roleplay paradigm. Replika (Maples et al. 2024) and the open-source mental-health LLMs (MentaLLaMA, SoulChat, EmoLLM, and adjacent work in Yang et al. 2024) ship with varying levels of safety scaffolding but no published constraint that prevents stripping it on fork. The fork-removes-safety failure mode is documented. In mid-2023, the NEDA helpline replaced human counselors with a generative-augmented "Tessa" chatbot that produced overt eating-disorder-promoting advice, including 500–1,000 calorie/day deficits recommended to ED-recovery users. After Sharon Maxwell's public testing surfaced the harm on May 30 2023, NEDA suspended Tessa within 24 hours, on May 31 2023, before the June 1 scheduled rollout that would have replaced the helpline staff.

The regulatory environment shifted accordingly through 2025. The FTC issued 6(b) orders in September to Meta, OpenAI, Character Technologies, Alphabet/Google, and others on chatbot companions and youth mental health. The FDA Digital Health Advisory Committee considered

generative-AI mental-health devices on November 6, 2025 and recommended Predetermined Change Control Plans, performance monitoring, and double-blind RCTs. The APA issued a Health Advisory the same month. In *Garcia v. Character Technologies* (M.D. Fla. 6:24-cv-01903), Judge Anne Conway's May 20 2025 ruling denied Defendants' motion to dismiss on First-Amendment-defense grounds, allowing negligence, product-liability, wrongful-death, and FDUTPA claims to proceed; only intentional-infliction-of-emotional-distress was dismissed. The case settled on January 7 2026 with terms undisclosed, covering Garcia and a related series of suits in Colorado, New York, and Texas. The settlement is consistent with elevated litigation-risk exposure under Conway's Order, but defendants settle for many reasons that are not equivalent to conceding the substantive question (cost, distraction, discovery exposure), so the Conway ruling rather than the settlement is the substantive precedent on whether chatbot output is shielded from tort liability by the First Amendment. Any deployed AI mental-health system shipping in 2026 operates under expanded operator-liability exposure with no clear regulatory pathway.

What is missing from the published literature is a governance pattern that is inspectable from outside, constraining when forked, and operable by a single developer or small team without commercial revenue. This paper proposes and operationalizes such a pattern.

1.2 Three-tool package

We propose three tools that together compose what we call the minimum viable safety scaffolding for a deployed public-good AI mental-health system. None of the tools is novel individually; each has substantial prior art cited below. The contribution is the combination applied to this deployment context, with documented search methodology supporting the prior-art claim.

1. A restricted-use license on system-prompt text (`LICENSE-PROMPT.md`) that requires named-section retention and reviewer-of-record substitution for derived deployments (§2.1).
2. A self-imposed deployment-pause posture with auditable trigger conditions, named adjudicators per trigger, and an author-unavailability dead-man switch (§2.2).
3. A CI-enforced rule-coverage check mapping registered safety rules to behavioral test assertions (§2.4).

Each tool addresses a different governance surface. The license addresses derivative deployments, since otherwise forks can silently strip safety-critical sections. The deployment-pause posture addresses the original deployment, since the operator's commitment to pause under named conditions is otherwise unenforceable from outside. The rule-coverage CI addresses both, by providing the deterministic safety-rule signal that the pause posture's rule-coverage trigger consumes. The three components do not form a closed enforcement loop. The CI feeds the original deployment's pause posture but does not constrain forks; the license constrains forks but does not gate the original. Closing the loop fully (for instance, by requiring under LICENSE-PROMPT that forks ship with an equivalent CI configured against their derived spec) is one direction of future-work refinement we flag in §6.2 and do not claim as part of this paper's package. What we offer is three independent tools designed to compose against the gap each was built for, with one direction of integration in place (CI → pause) and one acknowledged gap (CI → license).

1.3 What this paper is and is not

This paper is a governance-pattern proposal with one reference implementation and a documented prior-art search. The reference implementation is Stay: two of three components are production-active, the third is API-validated pending external-scheduler wiring. The paper is not a clinical-evaluation paper (Stay has no published RCT and is not FDA-cleared), not a regulatory-compliance pathway (the deployment is positioned outside regulated medical-device categories, which §5 lists as a limitation), and not legal advice (LICENSE-PROMPT has not been tested in court).

The companion paper contains the empirical work this paper defers: a four-utterance-class anti-pattern catalog, a measurement-validity reading of the VERA-MH benchmark, and a lookup-table-effect self-audit applied to Stay's own v0.7 specification. Citations to those findings appear here as motivation; the empirical claims are made there. Both papers and the reference implementation are at https://github.com/wudaming00/stay.

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2 · The three-tool package

The package's three components target different governance surfaces: the license addresses *derivative* deployments (so forks cannot silently strip safety-critical sections); the deployment-pause posture addresses the *original* deployment (so the operator's commitment to pause under named conditions is externally verifiable); the rule-coverage CI addresses *both* by providing the deterministic safety-rule signal the pause posture's rule-coverage trigger consumes. We describe each in turn.

2.1 LICENSE-PROMPT: use-restricted system prompt sections

A deployed AI mental-health system's safety behavior is encoded primarily in its system prompt: crisis-trigger logic, forbidden utterances, imminent-risk handling, referral resources. MIT-style permissive release lets forks both improve the prompt and strip safety-critical sections. A README note asking forks not to strip is unenforceable; a copyright-license-level constraint is.

The Stay System Prompt License v1 (`LICENSE-PROMPT.md`) governs the file `src/lib/system-prompt.ts` and substantially-derivative prompts. It grants broad redistribution, modification, and commercial-use rights with carve-outs for four named sections. §1.a is the imminent-risk SOP (method-driven persuasion plus means restriction plus companion-during-call). §1.b is the leverage-prevention rule (no weaponization of named reasons-for-living). §1.c is the no-third-party-characterization rule (no escalating user's framing of partner, parent, or friend). §1.d is the companion-during-call requirement, designated *preserved-pending-clinical-validation*. A derived deployment modifying any of these sections must publicly name a clinical reviewer-of-record (defined as holding active state licensure in a clinical mental-health discipline with crisis-intervention experience), document the modification, and publish the modified text. Derived deployments that retain these sections inherit the deploy-pause posture described in §2.2.

The fourth section requires a different framing than the others. Companion-during-call is grounded in crisis-line warm-handoff literature (Gould et al. 2007; Mishara & Daigle on Samaritans) but has not been validated for AI-companion-during-call interactions specifically. Locking an under-evidenced protocol via copyright would be paradoxical if read as "established practice." We read it instead as a protection against silent stripping, with license substitution

available under a transparency requirement: the reviewer-of-record path in LICENSE-PROMPT §2 is the explicit update channel for substantive modifications. A fork engaging a clinical reviewer-of-record may modify §1.d freely under §2 terms. What the license blocks is silent removal by forks that have done no clinical evaluation. The same asymmetric standard applies to the original Stay deployment. The §2.2 clinician-review trigger commits Stay itself to obtaining the same review on a public 2026-06-30 deadline, with deployment-pause consequences if missed. Status (and any post-deadline update) is in the project's public clinician-audit and deployment-conditions registries, which supersede this paper's date-stamped account at any point past the deadline.

LICENSE-PROMPT combines three prior patterns. From the Hippocratic License v3 (HL3, EthicalSource), it inherits the use-restriction approach; HL3 has over 300 adopters per firstdonoharm.dev, none of them deployed AI mental-health systems. From the OpenRAIL-M family (BigScience BLOOM, Stable Diffusion v2), it inherits the behavioral-constraint approach, although OpenRAIL applies to model weights and LICENSE-PROMPT applies to prompt text. From Mozilla MPL v2, it inherits file-level copyleft, although MPL targets the file as a whole and LICENSE-PROMPT targets named sections within a file. The combination of section-level retention, a reviewer-of-record substitution clause, and application to a deployed AI mental-health system's prompt is, by our search in §4, not previously published.

LICENSE-PROMPT has substantial limitations, listed in §5. Enforceability has not been tested in court, the legal quality has not been reviewed by counsel, and the reviewer-of-record substitution mechanism creates a gating step that some adopters may treat as friction rather than feature. A lawyer's review is the most important external obligation on the tool.

2.2 Deployment-pause posture with auditable triggers

The phrase "we will pause the deployment if something goes wrong" is unenforceable as a public commitment because both "something goes wrong" and "we will pause" are operator-discretionary. A monitoring-without-trigger commitment cannot be audited from outside, and neither can a trigger that lacks named adjudicators or deadlines. The pause posture has to be specific enough that an external reviewer can check whether the operator actually paused when they said they would.

Stay commits to four trigger conditions. Each has a named adjudicator, a defined adjudication artifact, and a time bound.

The *clinician-review trigger* fires if no licensed clinician has provided written review of at least one of the imminent-risk SOP, the companion-during-call protocol, the leverage-prevention rule, or the per-population SOPs by 2026-06-30. On firing, the companion-during-call feature and the method-driven imminent-persuasion SOP move behind a default-off feature flag, and the deployment's positioning text changes from product-mode to "research preview, not for use during active distress." The adjudicator is the author. The clinician sign-off, or its absence, is committed to the project's public `clinician-audit.md` registry by 2026-07-07 as the auditable artifact.

The *sentinel adverse event trigger* fires on a verified credible report of suspected user death, near-death, or severe iatrogenic harm. Verification requires both a reporting party with confirmable identity (licensed clinician, documented family member, law enforcement, or masthead media organization) and independent confirmation of the specific circumstance. The deployment enters

referral-only mode within 24 hours of verification, pending external review. The pause is invoked manually by setting the heartbeat-store gate to `expired` (the same substrate the dead-man switch uses; see §2.3); it is not an automated pipeline. The adjudicator is the author for the manual pause, and an external clinician reviewer (named at the time, drawn from the clinician-audit registry) for the resume decision. Receipt-side automation, a soft-pause-banner on plausibly-credible reports prior to verification with rate-limiting and de-duplication against griefer-spam, is a v0.9 milestone documented in the project's public deployment-conditions registry. The current commitment is the manual verified-hard-pause path. Single-author adjudication is the residual exposure; multi-party HMAC threshold signing once a clinician-of-record is engaged is the natural v1 hardening (cross-referenced from §2.3).

The *rule-coverage trigger* fires if `npm run check-rule-coverage` shows critical-rule coverage gaps unresolved for more than 30 days. The deployment is paused until coverage is restored. The adjudicator is the CI check itself, which is deterministic. Enforcement is live as merge-blocking branch protection via the project's GitHub Actions workflow at `.github/workflows/check-rule-coverage.yml`. A runtime kill-switch (a `DEPLOY_GATE_OPEN` env var consumed in the chat route, which would immediately move the deployed instance to referral-only mode on CI failure rather than only blocking new merges) is a v0.9 milestone documented in the deployment-conditions registry.

The *legal-demand trigger* fires on a verified credible legal demand related to user safety: a subpoena or regulatory inquiry from the FTC, FDA, or a state AG (verified via PACER, the regulator's docket, or direct phone confirmation to the issuing office); a tort complaint filed in court (PACER-checkable); or a cease-and-desist from counsel-of-record (firm verified in the state bar registry). On verification, the deployment enters referral-only mode within 24 hours, pending counsel review. The verification window is 72 hours from receipt. Same-day verification by a single uninsured author would frequently produce false-positive pauses on malformed or fraudulent demands. Unverified-pending-72h demands get a soft-pause warning banner without the hard pause. The adjudicator is the author for the post-verification pause and counsel for the resume decision.

Each trigger is committed in the project's public `deployment-conditions.md` registry along with the named adjudicators, deadlines, and any past invocations. A skeptical reviewer can verify operator compliance on any date. The closest published precedents are research-trial deployment protocols, including Therabot's IRB-supervised pause-on-flag practice and Wysa's clinical advisory board, and the safety-cases tradition in safety-critical software engineering (DO-178C, ARP4761), in which named conditions and adjudicators are required for certification but not for ordinary operation. We adapt the named-conditions-and-adjudicators pattern from safety-cases to ordinary operator-self-restraint, without the certification scaffolding. To our knowledge, no published deployed AI mental-health system commits to a public, auditable, trigger-and-adjudicator deployment-pause posture of this form.

2.3 Author-unavailability dead-man switch

The deployment-pause posture in §2.2 names the author as the adjudicator on two of the four triggers. This creates a single point of failure. If the author is unavailable through illness, travel, or abandonment, the deployment continues operating without the adjudication mechanism that justifies its operation.

Our first-pass design used a scheduled GitHub Actions workflow checking last-commit-age.

GitHub auto-disables scheduled workflows in public repos after 60 days of inactivity, which defeats the trigger precisely when sustained inactivity is the firing condition. We flag this pitfall explicitly so other AI mental-health projects do not repeat it.

The replacement design uses signal-loss-default-deny with a calibrated grace window. An external cron-as-a-service provider (cron-job.org or equivalent), independent of the project repo and the deployment platform, fires a daily HMAC-SHA256-signed heartbeat to the deployment's `/api/admin/heartbeat` endpoint. The endpoint validates a ±10-minute drift window for replay protection and uses `timingSafeEqual` for signature comparison. On each request, the deployment computes `now - last_heartbeat` and returns one of three statuses: `healthy` (≤48 hours), `warning` (48 hours to 7 days, with a banner added to the response), or `expired` (more than 7 days, request returns referral-only). The 7-day cutoff is calibrated to single-author normal life. Weekend illness, family emergency, and international travel routinely produce 48–72-hour gaps, and a tighter cutoff would fail-closed a safety-critical service on the typical case. Beyond 7 days, the deployment fails closed regardless of why the heartbeat stopped. Implementation: see the `src/lib/heartbeat-store.ts` module (validity and warning windows, status function), the `src/app/api/admin/heartbeat/route.ts` endpoint (consumer with replay protection), and the `src/app/api/chat/route.ts` route (gate consumer) at https://github.com/wudaming00/stay.

The mechanism has substantial limits on the infrastructure side. Persistence: `last_heartbeat_at` is module-level state, so Vercel serverless cold-starts reset to "unknown" (which the chat-route consumer treats as healthy by design; the rationale is in the source comments). Production-grade persistence via Vercel KV or Edge Config is a v0.9 milestone. Independence is only partial: the author configures cron-job.org, holds the HMAC secret, and can disable the cron from the same machine that writes the system prompt. The redesign relocates the single point of failure from GitHub-Actions infrastructure to cron-job.org infrastructure rather than removing it. The mechanism defends against unavailability, not against adversarial action. Multi-party HMAC threshold signing across the author, a clinician reviewer, and a neutral third party is the natural v1 hardening once a clinician-of-record is engaged. Provider failures, network partitions, and secret compromise are each individually well-mitigated by the 7-day grace plus fail-closed semantic. In aggregate, the mechanism is a default-safe state under ordinary unavailability, not a security control.

The clinical-side limit is sharper. Fail-closed-during-crisis versus degraded-availability is a live clinical question: a user in `expired` mode at 3 AM receives a referral-only response naming 988, Crisis Text Line, and 911 instead of Stay's companion protocol. The 7-day grace plus 48-hour warning configuration was chosen over a tighter 48-hour pause precisely to minimize this exposure on the typical case. The right calibration on the boundary case is unsettled pending clinical reviewer-of-record input.

Deployed-system dead-man switches in AI mental health, defaulting to referral-only on operator-unavailability, are not standard practice and we have not located prior published examples. The contribution here is the framing of "fail closed on signal loss" as the correct default for safety-critical deployed AI, the GitHub-Actions self-disabling finding as transferable engineering knowledge, and the integration with §2.2's trigger structure.

2.4 Rule-coverage CI

Behavioral testing of LLM systems with templated assertions (CheckList; Ribeiro et al. 2020)

catches regressions on assertions that were written, but does not catch the case where a safety rule was added to the system prompt but no test was written for it. The rule-test mismatch is the main maintenance hazard in deployed safety-critical conversational systems: a rule that nominally exists in the prompt but has no test will silently regress on prompt-update or model-update.

A hand-maintained `REQUIRED_RULES` registry enumerates the safety rules that must have at least one critical-severity behavioral assertion testing them. Each scenario in the test suite (61 scenarios as of v0.8) contains assertions; each critical-severity assertion carries a `rule:` field naming which `REQUIRED_RULES` entry it tests. The `npm run check-rule-coverage` script enforces, for every entry in `REQUIRED_RULES`, that at least one critical assertion references it. The check runs in CI and fails the build if coverage drops to zero on any rule. Adding a new rule to `REQUIRED_RULES` without writing an assertion fails CI immediately, as does removing all assertions for a rule or renaming an assertion's `rule:` field without updating the registry.

What the check does not catch is the converse: adding a new rule to the system prompt without registering it in `REQUIRED_RULES` is silent. The registry is hand-maintained, and there is no automated check that rules-in-prompt is a subset of rules-in-registry. Beyond this omission gap, the v0.8 spec contains five items deliberately not registered: three named safety rules (`inviolable.3` "never assert another's intent," `inviolable.7` "no comparison to others, even favorably," and `protocol.8` "rupture/repair check-in"), and two v0.8 agency-trajectory functions (`agency.memory` "long memory plus pattern surface" and `agency.logger` "structured journaling"). These five require test infrastructure not yet built: long-conversation scenarios for `protocol.8` and the rupture-check, and a multi-session harness for the two agency-affordance functions. They are documented as TODO with prefix-comments in the registry source. The CI check on the registered rules passes; the registry-omission gap is openly disclosed both in the registry source comments and in the project's `deployment-conditions.md` registry. Semantic adequacy is also not checked: an assertion can pass while the rule it nominally tests is misimplemented in the prompt.

The methodology root is CheckList's MFT/INV/DIR three-test-type taxonomy (Ribeiro et al. 2020). The conceptual root for rule-to-test mapping is bidirectional traceability in safety-critical software (DO-178C, ISO 26262, FDA SaMD Premarket Submissions guidance). We borrow loosely from both: templated behavioral assertions from CheckList, and registry-driven coverage checking from safety-critical traceability. This is a v1 partial implementation; it does not approach the verification depth those standards demand. CheckList has been applied to many domains, but we have not located a published application to clinical safety rules in deployed AI mental-health systems with registry-driven coverage gating in CI. Adjacent work in this space (MentaLLaMA, PsyEval) tests classifier accuracy or QA quality; rule-coverage as a deployment-gating CI check is, to our knowledge, new in this domain.

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3 · Stay as a reference implementation

3.1 Architecture summary

Stay is a Next.js 15 web application backed by `claude-sonnet-4-5-20250929` via Anthropic's API. The system prompt (~6,500 tokens at v0.8) is injected on every conversation. User conversations are stored in IndexedDB, encrypted with AES-GCM-256 using a key the

browser holds; the server cannot read conversation content. Default retention is 90 days, user-configurable. Crisis tooling, panic-phrase quick-exit, and Stanley-Brown stabilization-window safety planning are implemented in client-side React components.

Concrete file references for this paper's contributions are in the next three subsections; all paths are under https://github.com/wudaming00/stay.

3.2 LICENSE-PROMPT in practice

The system prompt file (`src/lib/system-prompt.ts`) opens with a TypeScript-comment license header referencing `LICENSE-PROMPT.md`. The four named protected sections inside the prompt's template-literal body are explicitly delimited by HTML-style comments visible in the raw text (`<!-- PROTECTED SECTION: imminent-risk SOP — see LICENSE-PROMPT.md §1.a -->` … `<!-- /PROTECTED SECTION: imminent-risk SOP -->`). The HTML-comment form is used because it survives unchanged through markdown rendering by clients while remaining greppable from the file system; TypeScript-comment markers (`//`) would not work inside the template literal. A derived fork that removes these comment markers removes the in-text indication of which sections are protected but the license still applies; the canonical mapping of sections to license clauses lives in `LICENSE-PROMPT.md`, not in the prompt comments.

The reviewer-of-record substitution clause has not been invoked at the time of writing — no fork has been documented to the project. The clause's first test will be when one occurs.

3.3 Deploy-pause triggers wired up

The rule-coverage trigger is currently enforced via GitHub branch protection: the `.github/workflows/check-rule-coverage.yml` workflow runs `npm run check-rule-coverage` on every push and pull request to main, and a coverage-gap failure blocks merge. A *runtime* kill-switch — a `DEPLOY_GATE_OPEN` env var consumed in `src/app/api/chat/route.ts` that would pause the running deployment immediately on CI failure (rather than only blocking new merges) — is a v0.9 milestone documented in the deployment-conditions registry and is not yet wired. The dead-man-switch heartbeat gate is live as a runtime gate; the rule-coverage gate is currently merge-time only.

The dead-man switch is implemented as the signal-loss-default-deny-with-grace-window pattern described in §2.3. The deployment-side consumer at `src/app/api/admin/heartbeat/route.ts` accepts POSTs of `{timestamp, signature}` JSON bodies, verifies HMAC-SHA256 against the `STAY_HEARTBEAT_SECRET` env var with a ±10-minute drift window for replay protection and `timingSafeEqual` for signature comparison, and writes the timestamp to the heartbeat store at `src/lib/heartbeat-store.ts`. The store exports `getHeartbeatStatus()` which returns `healthy` / `warning` / `expired` / `unknown` per the 7-day validity / 48h warning windows. The chat route at `src/app/api/chat/route.ts` calls `getHeartbeatStatus()` immediately after the daily-cap check and (a) returns a referral-only 503 response when status is `expired`, and (b) emits a `heartbeat_status` SSE event with a warning-banner payload when status is `warning` so the client renders the operator-may-be-unavailable notice above the conversation. The external scheduler component (cron-job.org or equivalent firing the daily heartbeat against the deployment URL) is the cron-side wiring task and is not part of the project repository.

Current operational status, per the deployment-conditions registry: the deployment-side endpoint, store, gate, and warning-banner SSE emission are implemented and validated. The external scheduler is not yet configured at the time of this paper's writing — the mechanism is therefore in *demo / API-validated* state, not in *production-active* state. Until the cron is configured, the deployment runs in `unknown` heartbeat status (treated as healthy per the cold-start semantic). We surface this gap explicitly because a hostile reviewer would otherwise catch it by clicking through to the deployment-conditions registry, which is the public-auditable artifact this paper deliberately points at.

The clinician-review and sentinel-event triggers are operator-discretionary at the moment of trigger and committed-to-repo as auditable artifacts after.

3.4 The 61-scenario test suite + rule-coverage check

The test suite (`scripts/scenarios/`) contains 61 scenarios across 13 categories (suicide, DV, leverage, trauma, psychosis-mania, OCD, ED, substance, threats, caregiver, daily, calibration, parasocial). Each scenario contains a sequence of user turns and a list of behavioral assertions. Six assertion kinds are supported: `must_call_tool`, `must_not_call_tool`, `must_match` (regex pattern present), `must_not_match` (pattern absent), `max_occurrences` (pattern bounded), `judge` (LLM-as-judge evaluates yes/no proposition).

The rule-coverage script at `scripts/check-rule-coverage.ts` enforces the rule-↔-assertion coverage. The current `REQUIRED_RULES` registry contains 33 rules; the script reports `33 covered, 0 uncovered` at v0.8. Five spec items are deliberately not registered (`inviolable.3`, `inviolable.7`, `protocol.8`, `agency.memory`, `agency.logger`) pending test infrastructure; their TODO status is documented in the registry source comments.

The full suite (judges enabled, 3-concurrency) runs in 12-18 minutes wall-clock at $3-6 per pass on Sonnet 4.5. `--no-judge` mode for programmatic-only assertions runs at $0.50 per pass and is the mode CI uses.

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4 · Prior art and search methodology

4.1 Documented search

The §1.2 novelty claim depends on a defensible search. We searched (2026-04-15 to 2026-04-29): arXiv (cs.HC / cs.CY / cs.AI / cs.CL / cs.SE), ACM DL, IEEE Xplore, JMIR archives, PubMed, the Hippocratic License registry (firstdonoharm.dev/companies), the OpenRAIL registry, and GitHub topic searches (`mental-health`, `chatbot`, `safety`, `license`, `deployment-pause`). Search-term combinations: "AI/LLM/chatbot mental health" × "system prompt license / prompt restricted use / Hippocratic License / OpenRAIL / CheckList / deployment pause / dead-man switch / rule coverage / safety."

What we found: ~25 deployed AI mental-health systems were reviewed (Therabot, Wysa, Limbic Access, Replika, Character.AI, Woebot, Pi, Earkick, Mindstrong, Tessa-NEDA, MentaLLaMA, SoulChat, EmoLLM, ChatPsychiatrist, Mental-LLM, X2AI Tess, Youper, Yana, Pixel.bot, Talkspace AI, BetterHelp AI, Lasting AI, Sintelly, Sonia, Wysa NHS); ~7 have peer-reviewed evaluation. Of all 25, *zero* publish their full system prompt, *zero* publish trigger-

condition + adjudicator commitments, and *zero* publish a rule-coverage CI artifact tied to a registered rule set. HL3 adopters span analytics / infrastructure / developer-tools but none deployed AI mental-health; OpenRAIL is applied to model weights, not prompts; ~5 papers apply CheckList-family methodology to LLM mental-health evaluation (MentaLLaMA eval, PsyEval, MHSafeEval, FAITA-MH, VERA-MH), none with a CI-gated rule-coverage registry. Research-trial protocols (Therabot IRB pause-on-flag) and regulated medical-device safety cases (DO-178C, ARP4761) embody auditable-trigger patterns privately, not as public commitments for ordinary operation.

Combination claim: across all corpora, zero published deployed AI mental-health systems combine all three tools. The novelty claim is made against an empty intersection.

What we cannot rule out: internal governance documents at closed-commercial deployments (Anthropic, OpenAI, Replika, Character.AI) are not public; HIMSS / Becker's-style trade-press venues are not exhaustively searched. We treat the claim as "no published deployed-system equivalent," not "no equivalent exists anywhere," and invite correction.

4.2 What we explicitly do not claim is novel

HL3-style use-restriction (since 2020); OpenRAIL behavioral constraint on ML weights (since 2022); MPL file-level copyleft (decades); CheckList behavioral-testing methodology (Ribeiro et al. 2020 ACL Best Paper); bidirectional traceability in safety-critical software; auditable trigger conditions in regulated medical-device development; open-source AI mental-health systems (MentaLLaMA, SoulChat, ChatPsychiatrist, Mental-LLM). What is novel is the *combination* applied to a free, public-good, single-author AI mental-health deployment, with LICENSE-PROMPT targeting safety-critical sections within a system prompt over a vendor base model.

4.3 Adjacent work that could be confused with our claim

Wysa's UKCA Class IIa medical-device status is a regulatory pathway, not a governance package; closed-source, does not publish prompt / triggers / coverage check. Therabot's IRB-supervised trial (Heinz et al. 2025) had internal governance but is not publicly documented in transferable-pattern form; weights and prompt are not public. Anthropic's well-being interventions (Anthropic 2025) — sycophancy reduction, self-harm classifier, ThroughLine resource partnership — operate at the model-vendor / substrate level. Stay's three-tool package operates at the application-deployment level above the substrate; the two layers are complementary.

The Stade et al. (2024) READI framework for responsible LLM deployment in behavioral healthcare is a vocabulary and stage model, not a tool package. READI provides language for *what* deployment readiness looks like across its proposed stages; our package proposes *how* a public-good deployment can address some of those readiness criteria with concrete artifacts. We position the work as offering tooling that addresses parts of READI's governance vocabulary (specifically: the "responsible deployment", "ongoing monitoring", and "harm response" dimensions) without claiming our work constitutes READI-stage compliance — Stade et al.'s framework is more comprehensive than the slice our three-tool package addresses, and full READI compliance for a clinical-grade deployment would require additional artifacts (formal IRB protocol, pre-registered evaluation protocol, etc.) that Stay does not have. We treat the two as complementary at different layers of abstraction.

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5 · Limitations

The pattern's portability is asserted, not demonstrated. As of 2026-04-30, no external party has adopted the three-tool package or any of its components for an AI mental-health deployment of their own. LICENSE-PROMPT has not been tested in court and the legal text has not been reviewed by counsel. The file-level copyleft mechanism may interact poorly with derivative-work doctrine, and the reviewer-of-record-substitution clause may be challenged as unduly burdensome. Pro-bono review from open-source-license counsel (Software Freedom Conservancy, EFF, university clinics) is being actively sought.

The author is unincorporated and uninsured. There is no LLC, no errors-and-omissions or cyber-liability coverage, and no prior legal-demand experience. The deploy-pause legal-demand trigger described in §2.2 is responsible operator behavior, but it does not substitute for incorporation and counsel retention. Readers considering this pattern as a fork base should not infer that operating without these protections is recommended.

The rule-coverage CI is a v1 partial implementation. All 33 rules in the registry are covered, but five spec items are deliberately omitted from the registry pending test infrastructure (§2.4). The check is one-directional and does not verify semantic adequacy. Adverse-event monitoring is structurally limited by the local-only encrypted memory architecture. Defensible substitutes (anonymized rule-compliance telemetry in the chat route, opt-in outcome surveys, sentinel-event reporting, IRB-supervised research partnership) are substitutes, not equivalents. This is the central unsolved problem of the deployment.

Stay deploys to US users in English. The adjudicator names, the legal-demand-trigger agencies, and the LICENSE-PROMPT copyright-law references are all US-specific. International forks would need to remap each.

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6 · Discussion and invitation

6.1 What we are claiming

The three-tool package (LICENSE-PROMPT, an auditable deployment-pause posture with dead-man switch, and rule-coverage CI) is offered as one reference implementation of one possible governance pattern for deployed public-good AI mental-health systems. We claim its components have substantial prior art, and that the combination applied to this deployment context is, by our search, not previously published. We do not claim it is the only possible pattern, that it is optimal, that the reference implementation is fully production-active (the heartbeat external scheduler wiring is the open v0.9 task), or that adopting it suffices for clinical-grade safety. The pattern is the contribution; Stay is an instance.

6.2 What we want others to do

We invite others to fork or improve LICENSE-PROMPT. The drafting was done without legal counsel and the reviewer-of-record-substitution clause was designed without enforcement testing. A second draft from someone with open-source license expertise would strengthen the pattern significantly. We will not treat such a fork as competition; the closer the field gets to a standard use-restricted prompt license for safety-critical sections, the better for everyone

deploying.

We invite improvements to the deploy-pause trigger taxonomy. Our four triggers (clinician review deadline, sentinel adverse event, rule-coverage failure, legal demand) reflect the threats we currently see and are not exhaustive. Specific gaps that might warrant additional triggers include data-breach disclosure obligation, model-vendor change of terms (an Anthropic policy update affecting Stay's substrate, for example), and funding-runway exhaustion. The last is a real condition for any public-good deployment without commercial revenue; Woebot's June 2025 D2C shutdown is the recent precedent. Builders deploying under different threat models will need different triggers.

We invite adoption and improvement of the rule-coverage CI. The check is small, about 150 lines of TypeScript. Bidirectional checking (verifying that all rules-in-prompt have entries in `REQUIRED_RULES`) is a natural v2 extension that would close the silent-addition gap §2.4 acknowledged. Semantic-adequacy verification (whether the assertion actually tests what the rule means) is a much harder open problem and likely requires LLM-as-judge over rule-text and assertion-text pairs.

A note on the open loop between CI and license. §1.2 acknowledged that the rule-coverage CI does not currently constrain forks, and that we offer the three tools as composable but not closed-loop. A previous draft of this paper proposed amending LICENSE-PROMPT to require forks to ship an equivalent rule-coverage CI configured against their derived spec. We walked that proposal back. LICENSE-PROMPT's enforceability ceiling is already low (see §5 on the license being untested in court and possibly unenforceable in some jurisdictions), and adding a "must ship CI we approve of" clause would do three things, all bad: it would make the license materially more burdensome to comply with, it would require the licensor to evaluate whether each fork's CI is "equivalent" (a subjective standard that invites disputes), and it would increase the surface area for the license to be challenged as unduly restrictive. The right direction is to keep LICENSE-PROMPT minimal and defensible (named-section retention plus reviewer-of-record substitution, full stop) and to publish the rule-coverage CI script independently so adopters can choose to run it. Forks that want equivalence can adopt the script; forks that don't are still constrained by LICENSE-PROMPT's section-retention clause. The arrow we hoped to close (CI to license to fork) is genuinely missing, and we now treat it as a structural property of the package rather than a hole to plug.

A small registry of deployments operating under the package — the original Stay, plus any forks — would be valuable as an external accountability artifact. We will host such a registry at `docs/deployments-using-this-package.md` in the project repo, accept pull requests, and treat inclusion as a governance handshake (the fork enters the same auditable-pause posture as the original).

6.3 The scope honestly stated

This paper does not solve deployed AI mental-health safety. It does not approach the regulatory-evidence bar that FDA SaMD pathway demands, the clinical-evidence bar that JMIR Mental Health peer review demands, or the operational-monitoring bar that sentinel-event reporting demands. It offers one engineering-ethics scaffolding pattern that a public-good deployment can adopt as the *minimum viable* governance package while pursuing those higher bars in parallel. We are explicit that *minimum viable* is not *sufficient*: a fork that adopts this package and stops there is not, in our view, operating responsibly. The package is a floor, not a ceiling.

The deployment we built (Stay) is operating against this floor in part because the alternative — operating without any of these scaffolds — is what most public-good or grassroots AI mental-health deployments currently do. We propose a floor the field can hold each other to.

Acknowledgments

VERA-MH expert council (Bentley et al. 2026), Spring Health, and the Stade et al. (2024) authors for releasing or articulating the frameworks we position against. The Hippocratic License community (EthicalSource), the BigScience OpenRAIL drafters, and the Mozilla Foundation for the license-pattern lineage. Crisis-line counselors at 988, Crisis Text Line, and the National DV Hotline whose actual labor this scaffolding can only hope to support. The single-author operating posture this paper describes is provisional; clinical reviewer-of-record candidates and academic-partner expressions of interest at hello@thestay.app are welcomed.

Data and code availability

All artifacts referenced in this paper are public at https://github.com/wudaming00/stay.

- Reference implementation: the Stay project repo. Test suite, runner, and methodology infrastructure are released under MIT (`LICENSE`); the system prompt at `src/lib/system-prompt.ts` is released under the Stay System Prompt License v1 (`LICENSE-PROMPT.md`).
- Rule-coverage CI script: `scripts/check-rule-coverage.ts`.
- Rule-coverage CI workflow: `.github/workflows/check-rule-coverage.yml` (runs `npm run check-rule-coverage` on push and pull-request to main).
- Dead-man-switch implementation: `src/lib/heartbeat-store.ts` (validity / warning windows, status function), `src/app/api/admin/heartbeat/route.ts` (HMAC-SHA256 consumer with replay-protection drift window), `src/app/api/chat/route.ts` (status-gate consumer returning referral-only on expired).
- Deployment-conditions registry (public-auditable trigger conditions and any past invocations): `docs/deployment-conditions.md`.
- Clinician-audit worksheet and outreach queue: `docs/clinician-audit.md`.
- Companion paper on the four-utterance-class anti-pattern catalog and VERA-MH measurement-validity reading: `docs/paper-B-antipattern-catalog-draft.md` in the same repository.

Conflicts of interest

The author is the sole developer of Stay (free at thestay.app, no revenue, no advertising, funded out of pocket; not incorporated; uninsured). Stay has no commercial-entity backing, for-profit operator, or active accelerator / venture application. The author has unrelated commercial projects in real-estate analytics SaaS and voice-AI tooling (bootstrapped, no VC, no active institutional-investor relationship), none in mental-health, clinical AI, AI safety governance, or any adjacent domain where this paper's contribution might benefit reputationally. The argument of this paper does not depend on Stay being well-implemented; the package is offered as a pattern to fork or reject. Stay's free, public-good positioning is committed in `LICENSE-PROMPT.md`.

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